Day 1 Exercises

Foundations, Earth Engine, and vibe coding. Work through these as you go — Part A during the lecture, Parts B and C during the labs, Part D at home.

A Concepts — do these before you touch a computer

A1

A Sentinel-2 pixel covers 10 × 10 m on the ground. You are asked to map individual trees in a teak plantation where trees are planted 4 m apart. Can you do it with Sentinel-2? Explain in one sentence.

A2

Fill in the NDVI value you would expect for each surface.

SURFACE	EXPECTED NDVI	WHY
A reservoir		
Dense evergreen forest		
A harvested rice field		
A concrete car park		

A3

Your NDVI map for Chiang Mai has a maximum value of **2,847**. What went wrong, and what single line of code fixes it?

A4

Why do we build a **median composite** from three months of images instead of just picking the one clearest day?

A5

You need the total forest area of Kanchanaburi province. Which `scale` would you pass to `reduceRegion`, and why not a smaller one?

- ☐ `scale=10` — native Sentinel-2 resolution
- ☐ `scale=100`
- ☐ `scale=1000`

B The hallucination drill

DO NOT RUN THIS CODE

Below are four snippets an AI assistant produced. **Two of them contain an error that would stop them working**. Find them using the cheat sheet and the catalog — not by running them. This is the skill that matters most, and running the code is the slow way to acquire it.

B1

```
s2 = ee.ImageCollection('COPERNICUS/S2_SR_HARMONIZED') \
       .filterDate('2024-01-01', '2024-03-01') \
       .filterBounds(aoi)
ndvi = s2.median().select('NDVI')
```

Correct, or broken? If broken, what exactly is wrong?

B2

```
gfc = ee.Image('UMD/hansen/global_forest_change_2025_v1_13')
loss = gfc.select('lossyear').gt(0)
```

Correct, or broken? If broken, what exactly is wrong?

B3

```
biomass = ee.Image('NASA/ORNL/biomass_carbon_density/v1')
mean_agb = biomass.select('agb').reduceRegion(
    ee.Reducer.mean(), aoi, 300)
```

Correct, or broken? If broken, what exactly is wrong?

B4

```
wc = ee.Image('ESA/WorldCover/v200/2021').select('Map')
forest = wc.eq(10)
ha = ee.Image.pixelArea().divide(1e4).updateMask(forest) \
       .reduceRegion(ee.Reducer.sum(), aoi, 100, maxPixels=1e9)
```

Correct, or broken? If broken, what exactly is wrong?

C Lab work

C1 · Lab 1 — your own province

Run `lab1_first_map.ipynb` with `PROVINCE` set to a province you care about. Record:

QUANTITY	YOUR VALUE
Province name	
Area (ha)	
Number of Sentinel-2 scenes	
Mean NDVI	
Minimum NDVI	
Maximum NDVI	

Sniff test. Is your maximum NDVI less than or equal to 1? ☐ yes ☐ no — *if no, stop and fix it before continuing.*

C2 · Lab 2 — hectares

Run `lab2_landcover.ipynb` on the same province.

CLASS	HECTARES	SHARE	RANK
Tree cover			
Cropland			
Grassland			
Built-up			

Sniff test. Does your class total come close to the province area from C1? Write both numbers and the difference as a percentage:

C3 · The disagreement

Compare WorldCover tree cover with Dynamic World trees for your province.

PRODUCT	TREE COVER (HA)	SHARE OF PROVINCE
ESA WorldCover 2021		
Dynamic World		
Difference		

Give **two** reasons the products disagree. Do not write "one is wrong".

C4 · Write the sentence

State your forest area result the way a professional would — number, product, year.

D Homework

D1

Run both labs on a **second** province that you expect to be very different from your first — for example a heavily agricultural one in the central plain. Record the tree cover share of each and write one sentence explaining the difference in terms of what you know about the two places.

D2

Write a PROMPT-recipe prompt, from scratch, for a question we have *not* done: *"show me which parts of my province are above 1,000 m elevation."* You will need a dataset we have not used — search the Earth Engine catalog for a digital elevation model. Fill all six slots.

Hint: search the catalog for "SRTM". You do not have to run it — the exercise is the specification, not the result.